

Bayes' Classifier

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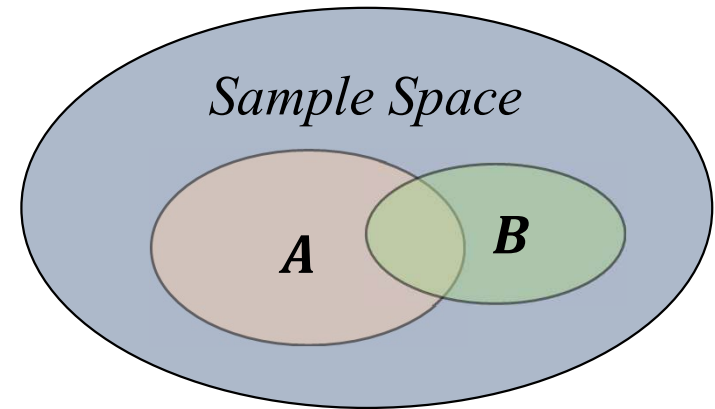
OUTLINE

- Revisiting Bayes' Rule
- Bayesian classification
- Bayes' Decision Rule
- Bayes' Decision Boundary
- Naïve Bayes classification algorithm
- Different variants of Naïve Bayes algorithm

Revisiting Bayes' Rule

Bayes' theorem is simply a consequence of conditional probabilities of two events A and B :

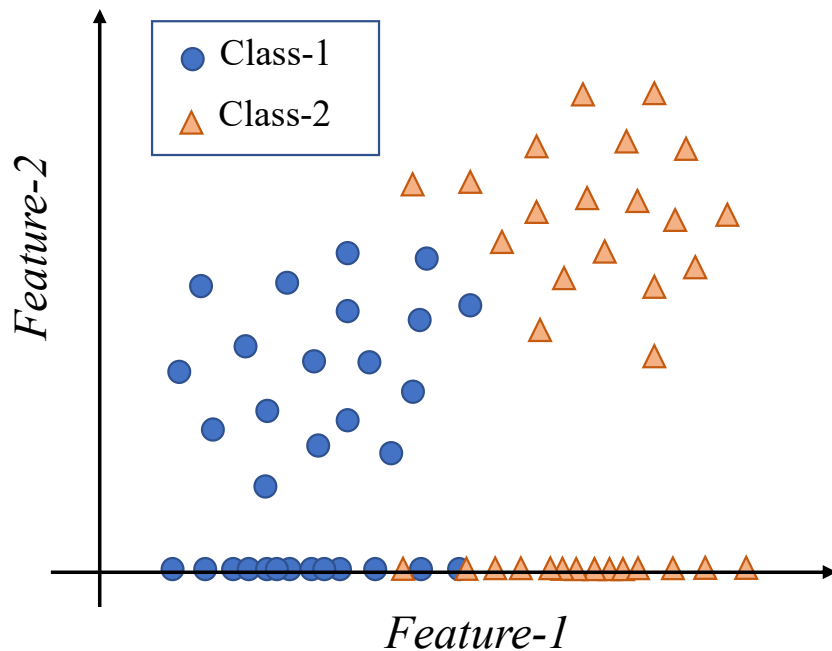
$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$



- $P(A)$: Prior Probability of event A
- $P(B|A)$: Likelihood of event B given event A
- $P(B)$: Evidence of event B
- $P(A|B)$: Posterior probability of event A given event B

Bayesian Classification

- The classification problem is posed in probabilistic terms.
- Create models for the distribution of objects of different classes.
- Probabilistic framework is used to make classification decisions.

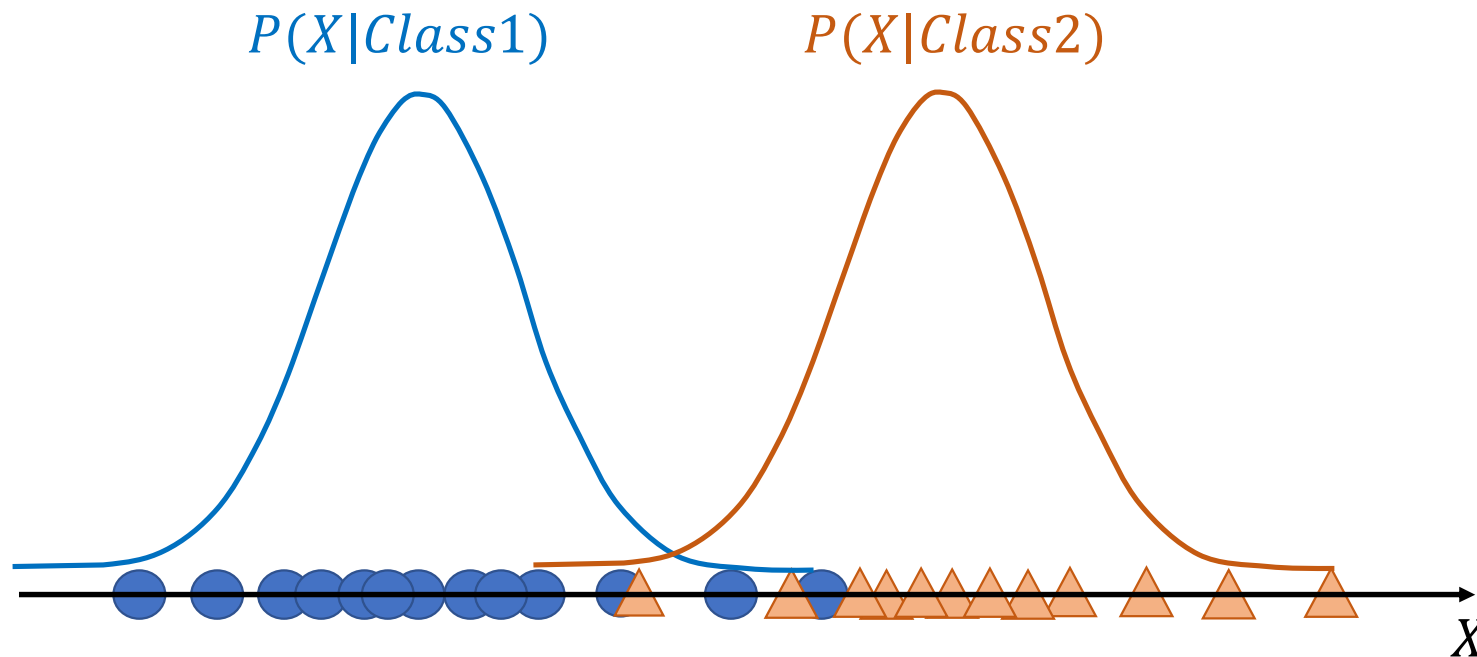


- Each object is usually associated with multiple features / predictors.
- We will look at the case of just one feature for now.
- We are now going to define two key concepts.
 - Likelihood or Class conditional probability distribution.
 - Prior probability of a class.

Bayesian Classification

- **Likelihood or Class Conditional Probability Distribution**

We model the likelihood function $P(X|Class\mathbf{k})$ using a probability distribution function.



In this case we have *modelled* these distributions using Gaussian / Normal distribution

Bayesian Classification

- **Prior probabilities of classes**
- We model *prior probabilities* to quantify the expected *a priori* chance of seeing a class.
- Let there are total m many training samples and out of which m_1 number of samples belong to Class-1 and $m - m_1 = m_2$ number of samples belong to Class-2
- Then prior probabilities are calculated as: $P(\text{Class1}) = \frac{m_1}{m}$ and $P(\text{Class2}) = \frac{m_2}{m}$
- Now we have *prior probabilities* of each class: $P(\text{Class1})$, $P(\text{Class2})$ and we also have models for the *likelihood functions* given each class. i.e. $P(X|\text{Class1})$ and $P(X|\text{Class2})$. Usually the likelihood function is modelled using some standard probability distribution function such as gaussian distribution.
- We want the *probability of the class given a pattern X*. i.e. $P(\text{Class1}|X)$ or $P(\text{Class2}|X)$

How do we get $P(\text{Class}|X)$ knowing $P(X|\text{Class})$ and $P(\text{Class})$?

Bayesian Classification

- We apply Bayes' rule to obtain $P(Class|X)$:

Posterior probability of
the class given the feature

Likelihood of the
feature given the class

Prior Probability of the
class

$$P(Class|X) = \frac{P(X|Class) P(Class)}{P(X)}$$

total probability of the
feature

Bayes' Decision Rule

- If we observe an object with feature X , how do we decide if the object is from Class-1?
- Bayes' decision rule is simply choose Class-1 if:

$$P(\text{Class1}|X) > P(\text{Class2}|X)$$

$$\text{or, } \frac{P(X|\text{Class1}) P(\text{Class1})}{P(X)} > \frac{P(X|\text{Class2}) P(\text{Class2})}{P(X)}$$

$$\text{or, } P(X|\text{Class1}) P(\text{Class1}) > P(X|\text{Class2}) P(\text{Class2})$$

$$\text{or, } \frac{P(X|\text{Class1}) P(\text{Class1})}{P(X|\text{Class2}) P(\text{Class2})} > 1$$

$$\text{or, } \log \left(\frac{P(X|\text{Class1}) P(\text{Class1})}{P(X|\text{Class2}) P(\text{Class2})} \right) > 0$$

$G(X)$

If $G(X) > 0$, we classify as Class-1, called MAP rule.

MAP: Maximum A Posteriori

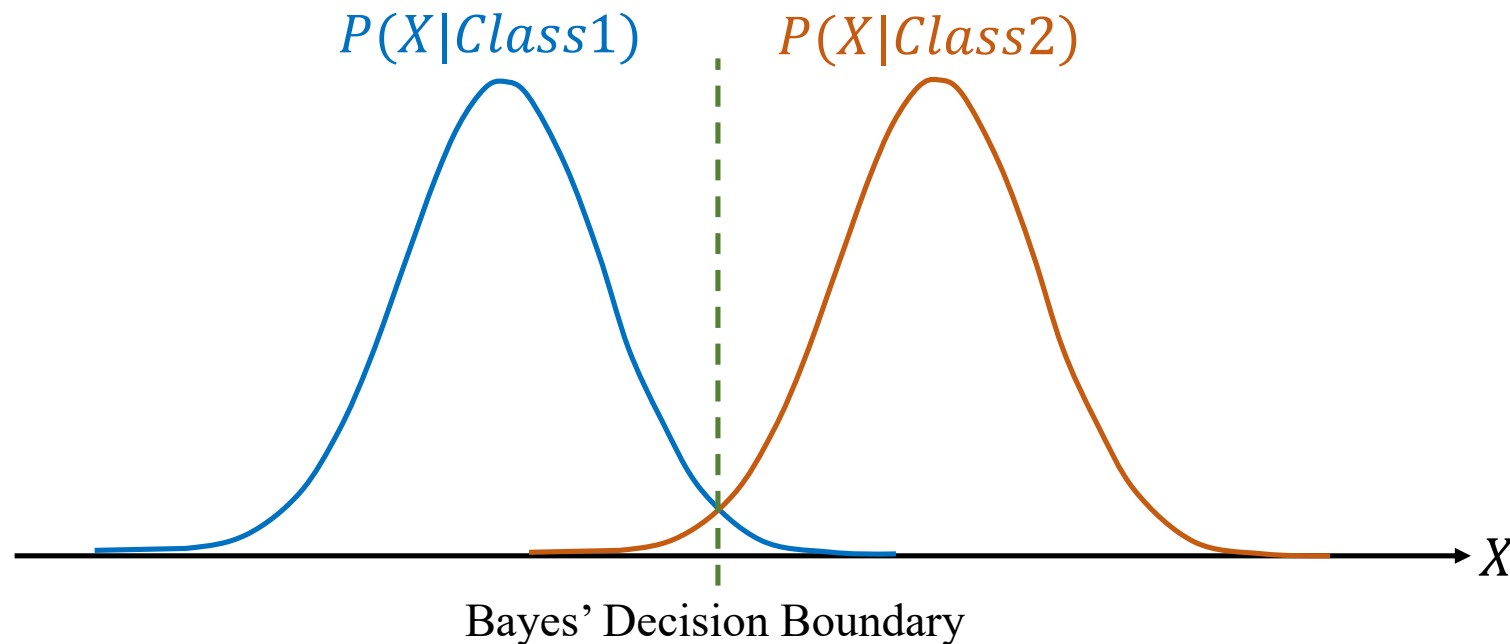
Bayes' Decision Boundary

- Bayes' decision boundary is obtained as:

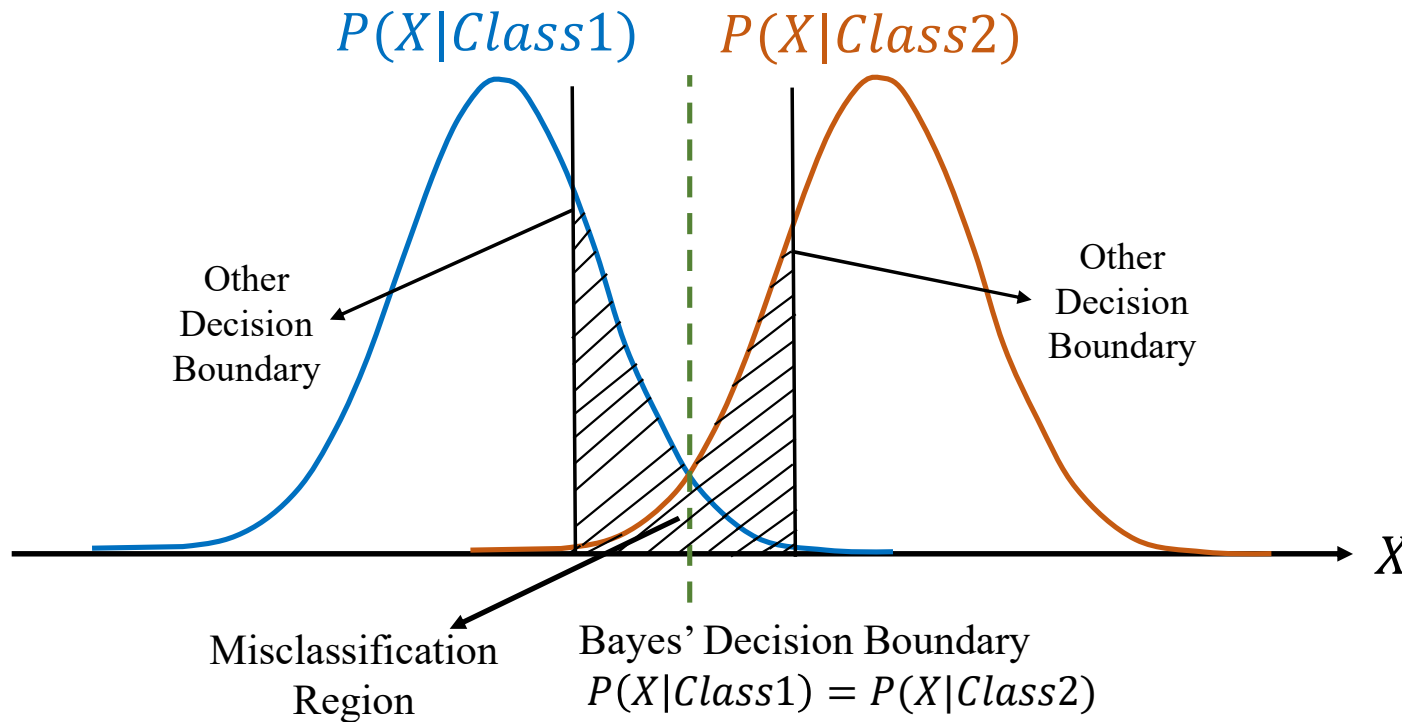
$$G(X) = 0 \Rightarrow P(X|Class1)P(Class1) = P(X|Class2)P(Class2)$$

- If we assume that prior probabilities of the classes are identical (balanced distr.) then:

$$P(X|Class1) = P(X|Class2)$$



Bayes' Decision Boundary



- Changing the decision boundary increases the misclassification error

- Bayes' decision boundary gives minimum misclassification error

For Multiple Features

- We have feature vector comprises of n features: $\vec{X} = [X_1, X_2, \dots, X_n]^T$
- Bayes' Classification: $P(C|\vec{X}) \propto P(\vec{X}|C) P(C)$
- Now: $P(\vec{X}|C) = P(X_1, X_2, \dots, X_n|C)$
- Difficulty: Learning the joint conditional probability $P(X_1, X_2, \dots, X_n|C)$

Naïve Bayes' Classification:

- Assumption that all input features are conditionally independent:

$$P(X_1, X_2, \dots, X_n|C) = P(X_1|C) P(X_2|C) \dots P(X_n|C) = \prod_{i=1}^n P(X_i|C)$$

- Maximum A Posteriori (MAP) rule: for $\vec{X} = [X_1, X_2, \dots, X_n]^T$ it belongs to class C_1 if:

$$[P(X_1|C_1) P(X_2|C_1) \dots P(X_n|C_1)]P(C_1) > [P(X_1|C_2) P(X_2|C_2) \dots P(X_n|C_2)]P(C_2)$$

How do we model the likelihood?

- The "magic" of Naive Bayes isn't in the algorithm's structure, but in the **likelihood distribution** you choose to fit *your specific type of data*.

Naïve Bayes variants	Input Data Type	Likelihood Distribution	Primary Use Case / Examples
Gaussian Naïve Bayes	Continuous (Real numbers)	Gaussian (Normal): Assumes a bell curve distribution	Predicting heart disease using continuous vitals like blood pressure, cholesterol levels (mg/dL), and age.
Bernoulli Naïve Bayes	Binary (0/1, No/Yes, False/True)	Benoulli : Focuses on whether a feature exist or not	Software bug type detection: The error log contains the word "Timeout" or not.
Categorical Naïve Bayes	Categorical (Fixed labels/ Discrete)	Categorical : For features with distinct, non-ordered categories.	Features like "Housing Status" (Rent, Own, Mortgaged) or "Marital Status" (Single, Married, Divorced).
Multinomial Naïve Bayes	Discrete Counts (Integers ≥ 0)	Multinomial : Based on the freq/repitition	Spam detection in Email: based on frequencies of the words appearing in both spam and non-spam emails.

Naïve Bayes' Classification

Advantages:

- Training is very fast; just require to consider each attribute in each class separately to model the likelihood function using some standard distribution (like Gaussian) and prior probabilities of each class.
- Test is straightforward: Calculate the likelihood function of each feature and multiply them with class a priori probabilities and then compare using MAP rule.
- Performance competitive to most of the state-of-the-art classifiers for simple use-cases.
- Many successful applications. E.g. Spam mail filtering.

Naïve Bayes' Classification

Relevant Issues:

■ Violation of Independence assumption:

- For many real world tasks, $P(X_1, X_2, \dots, X_n|C) \neq P(X_1|C) P(X_2|C) \dots P(X_n|C)$
- Nevertheless, Naïve Bayes' works surprisingly well even when independence assumption is violated.

■ Zero Conditional Probability Problem in case of non-continuous features:

- If no example contains the attribute value $X_j = a_{jk}$, then $\hat{P}(X_j = a_{jk} | C = C_i) = 0$
- In this circumstance, $\hat{P}(X_1|C_i) \dots \hat{P}(a_{jk}|C_i) \dots \hat{P}(X_n|C_i) = 0$ during test.
- For a remedy, conditional probability is calculated with a smoothing technique called *Laplace Smoothing*.

Thank You